

Global Supply Chain & Logistics Performance

Final Project Report

Student Information

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Track: Operations Analytics

Project Title: Global Supply Chain & Logistics Performance

1. Introduction

The project focuses on analyzing supply chain operations using the DataCo Supply Chain Dataset. The objective is to understand logistics activities, shipping performance, sales trends, customer behavior, and operational efficiency through data analytics.

2. Objectives

- Understand the structure of the supply chain dataset.
- Analyze logistics and shipping operations.
- Identify sales and profit trends.
- Examine delivery performance.
- Evaluate customer and market behavior.
- Generate business insights for supply chain optimization.

3. Tools and Technologies Used

Google Colab, Python, Pandas, NumPy, Matplotlib, Seaborn, GitHub.

4. Dataset Description

Dataset Used: DataCo Supply Chain Dataset.

The dataset contains information related to orders, customers, products, sales, profits, shipping modes, delivery status, markets, countries, and regions.

5. Project Workflow

Notebook 01: Data Loading

- Imported required Python libraries.
- Uploaded dataset files.
- Extracted ZIP files.
- Loaded CSV dataset.
- Verified dataset structure.
- Created dataset backup.

Notebook 02: Data Understanding

- Explored dataset features.
- Checked data types.
- Generated summary statistics.
- Identified missing values.
- Created initial visualizations.

Notebook 03: Data Cleaning

- Handled missing values.
- Removed duplicate records.
- Treated outliers.
- Saved cleaned dataset.

Notebook 04: Exploratory Data Analysis (EDA)

- Analyzed sales trends.
- Evaluated profit performance.
- Studied shipping modes.
- Examined delivery status.
- Compared regions and markets.
- Generated correlation analysis.

6. Key Findings

- Sales were concentrated in specific categories and regions.
- Profitability varied across products and regions.
- Certain shipping modes were used more frequently.
- Customer purchasing behavior showed identifiable trends.
- Multiple international markets contributed to growth.

7. Business Insights

- Improving delivery efficiency can enhance customer satisfaction.
- High-performing regions deserve additional business focus.
- Underperforming markets may require operational improvements.
- Data-driven inventory planning can reduce logistics costs.

8. Challenges Faced

- Handling missing values.
- Cleaning inconsistent records.
- Detecting and treating outliers.
- Understanding logistics-related variables.

9. Conclusion

The project successfully analyzed supply chain operations using the DataCo Supply Chain Dataset and generated valuable insights regarding sales performance, customer behavior, shipping operations, and market trends.

10. Future Scope

- Predict shipment delays using machine learning.
- Forecast future sales trends.
- Build interactive dashboards.
- Optimize inventory management.
- Implement advanced supply chain analytics solutions.

References

1. DataCo Supply Chain Dataset
2. Python Documentation
3. Pandas Documentation
4. NumPy Documentation
5. Matplotlib Documentation
6. Seaborn Documentation
7. Google Colab Documentation

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